

ST309 Project Report:

The Price of Performance: A Statistical Analysis of NBA Salary Drivers

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Introduction

Teams in the National Basketball Association (NBA) need to select their players, but the complicated salary cap regulations and subjective valuation methods often obscure the correlation between employee (basketball players) performance and income. And this project aims to address this challenge by creating a transparent quantitative framework to determine "fair market values" for current and potential NBA players. By analyzing a dataset comprising 7298 raw entries (from 1,658 players) and 31 performance variables from the 2010 – 2025 seasons, we aim to identify the specific on-court behaviors that drive salary outcomes.

Our investigation would prioritize interpretability over raw predictive power. We found the issue that a full model containing all 30 performance predictors could show a very severe multicollinearity issue because many performance statistics are highly correlated (e.g., "Twos_Made" & "Twos_Attempt"). But by applying Stepwise Selection and Generalized Variance Inflation Factor (GVIF) analysis, we refined our model. Although this reduced our Adjusted R^2 to roughly .60, it successfully improves the statistical noise problem, resulting in a set of actually relevant predictors that stakeholders can use for decision-making.

However, our findings also reveal a structural limitation in applying linear models to sports markets. Our final model retains a RMSE of approximately 5 million USD, which we think this variance is primarily driven by the "superstar premium phenomenon". Residual analysis indicates that while our model predicts the value of average NBA players relatively accurately, it consistently underestimates the wages of the top players. This could be due to the fact that we remove superstars & rookies in the training set, but it also suggests that NBA max contracts might also be determined by other "out-of-court" factors such as marketability, brand value, and ticket sales, which would lie beyond the scope of purely on-court statistics. Therefore, our research concludes that although performance metrics drive most of the NBA market, the valuation of superstars remains an exponentially growing phenomenon, and is difficult to measure using linear predictions with only on-court data.

Motivation

The NBA operates in a very competitive environment, from both performance and financial perspectives, contract decisions could be vital to any team's "success" or "failure". However, the determination of player salaries often appears to be a "black box" operation involving subjective judgment and emotional negotiation. Our project aims to address the core issue of identifying and quantifying the performance determinants of NBA player salaries. And this problem is best solved

using the data analysis approach because professional basketball exists under a "data-rich" environment: every player's action, from shots and assists to defense, is quantified as statistical measurements; on the other hand, their salaries provide an ideal numerical target.

The core motivation for this study is driven by the information asymmetry that plagues professional basketball. Players and agents aim to maximize career earnings within a limited professional "lifespan", while teams must allocate funds efficiently to remain competitive under a strict "salary cap". Historically, this tension was resolved through qualitative comparisons, where opaqueness and subjective judgements are involved. However, academic literature suggests these methods are flawed; Berri, Schmidt, and Brook (2006) argued in "The Wages of Wins" that decision-makers irrationally overvalue scoring volume while undervaluing efficiency and defense. Furthermore, the "Economics of Superstars" theory (Rosen, 1981) suggests that in global markets like the NBA, the relationship between talent and pay is often exponential (small differences in skill lead to exponential differences in reward), which introduces bias to qualitative judgements; therefore, the difficulty for both players and teams to negotiate a rational payment would significantly increase.

Our research aims to bridge this gap by establishing the historical correlation between NBA players' on-court performances and wages from 2010 to 2025. Unlike opaque machine learning models that lack explanations and accountability, we chose to use a more direct multiple linear regression approach to build a direct and interpretable framework. This allows users to understand the very specific mechanism of how individual statistics translate into earnings, providing a "fair market value" benchmark that provides information for different stakeholders in the industry.

Practical Utilities

The findings of this study offer direct practical applications for the league's primary actors. For NBA players and prospects (talented amateur players or potential NBA players), the model serves as a tool for rationalized career development. NBA players have limited training time and physical energy. And this project helps them to deal with such an optimization problem: which skills should they improve to earn more in their future careers? By examining our regression analysis, players can identify the specific marginal financial value of on-court performance metrics. The coefficient acts as a "market price tag" for each skill. This is particularly crucial for prospects, as they may still be at the early stages of their careers. Instead of attempting to be "well-rounded" in areas the market does not value, they can specialize in high-coefficient traits to maximize their initial contract value. Effectively, this transforms career planning from experience and intuition into statistical plans.

For General Managers and agents, the model provides a quantitative benchmark to improve existing information asymmetry. The model's utility relies on its residuals: the difference between a player's actual current salary and the model's predicted value. A player with a low actual salary but a high predicted one could be a representation of an "economically efficient" asset to the team. Vice versa, identifying players whose actual pay far exceeds their statistical production helps teams avoid long-term financial liabilities, allowing a more rationalized financial planning under the strict constraints of the salary cap. Ultimately, despite the "superstar premium" might always exist due to their "out-of-court value" in marketing, endorsement, etc., the majority of the NBA market follows a discoverable, data-driven logic that can be leveraged for competitive advantage, and our model provides information to help their decision-making.

Data Overview

Data source

Our dataset is attained from Kaggle, which contains NBA player-level information from 2010 to 2025 (<https://www.kaggle.com/datasets/ratin21/nba-player-stats-and-salaries-2010-2025>). It includes 7,298 observations of player salaries with 30 other season statistics (predictors), and these observations correspond to 1,658 unique players. The dataset includes the variables Player, Salary, Year, Pos, Age, and Team. Most other variables are basketball-specific abbreviations. To make the analysis easier to follow for non-basketball audiences, we renamed these abbreviations into more meaningful and interpretable variable names (Appendix A).

We group the variables into 2 categories:

- (i) player profile (Player, Age, Position, Team, Year); and
- (ii) performance measures (e.g., Games_Played, Minutes, Three_Attempts etc).

We treat Year, Pos, and Team as categorical variables (factors). We keep the remaining performance metrics as continuous variables.

Data cleaning and transformation

We inspected missingness across variables and found that missing values are relatively limited and concentrated in shooting-percentage measures (Figure 1), particularly Three_Percent (8.02%) and Free_Throws_Percent (2.77%), with the rest below 1%. Missing values all occur in percentage variables because attempts can be zero (division by zero). We code these missing percentages as 0

to represent “no attempts” and keep all observations.

	variable <chr>	na_count <int>	na_percent <dbl>
Three_Percent	Three_Percent	585	8.02
Free_Throws_Percent	Free_Throws_Percent	202	2.77
Two_Percent	Two_Percent	45	0.62
Field_Goal_Percent	Field_Goal_Percent	20	0.27
Effective_Field_Goal_Percent	Effective_Field_Goal_Percent	20	0.27
Player	Player	0	0.00

Figure 1

Aligning with our research goal: explaining and predicting Salary based on measurable performance and role-related factors. We keep Salary as the outcome variable and use the above predictors as candidate predictors. Since salaries are strongly right-skewed, the mean salary (6,483,666) is much higher than the median (3,010,634), and a small number of extreme contracts generate a long right tail. Therefore, we work primarily with $\log(\text{Salary})$ in our regression analysis. This transformation makes the distribution substantially closer to symmetric (Figure 2), which improves the plausibility of linear-model assumptions and stabilises variation.

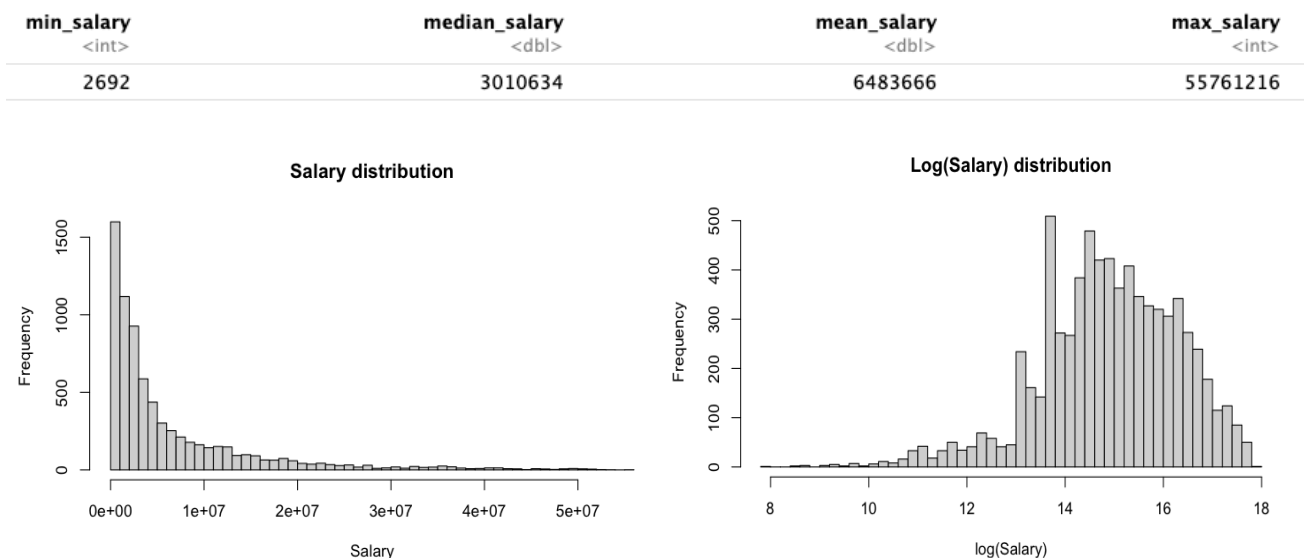


Figure 2

Exploratory Data Analysis

We carried out a simple EDA to understand the main patterns before fitting linear regression. We used box plots for categorical variables (Year, Pos, Team) and scatter plots for continuous variables, without labelling individual players to keep the figures readable(Appendix B, Appendix C). The box plots show that $\log(\text{Salary})$ shifts upward over time. Salary distributions also vary by position, while team differences are smaller compared with within-team variation.

The scatter plots show that $\log(\text{Salary})$ generally increases with playing time and production variables (e.g., Minutes and Points), but the relationship is not perfectly linear and there is

substantial spread. This suggests that single variables cannot explain salary well on their own, so a multivariable regression model is needed. We also observe signs of curvature in some relationships (diminishing returns at higher values), which motivates applying transformations to improve linear fit. In addition, the Age pattern appears non-linear, so we later allow a quadratic term (Age^2). Finally, a small number of extreme high-salary observations stand out from the main cloud, so we later check influential points using Cook's distance.

We also examined correlations among numeric predictors using a correlation heatmap (Figure 3). The heatmap shows strong clusters among predictors (e.g., minutes–shot attempts/makes–points), indicating potential multicollinearity, so we check this during model fitting (e.g., using VIF) when interpreting coefficients.

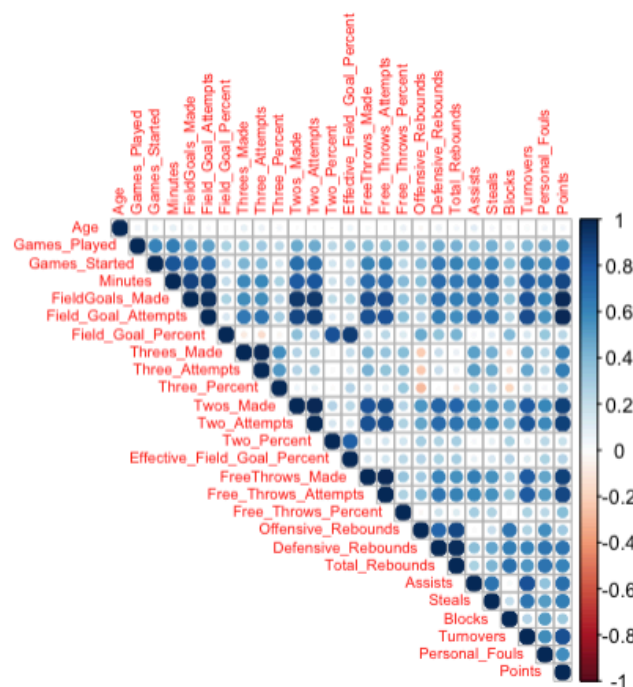


Figure 3

Regression Modelling & Analysis

Modelling setup and training data preparation

We split the data into a training set (95%) and a test set (5%) for out-of-sample evaluation. Since salaries are right-skewed, we model $\text{Logged_Salary} = \log(\text{Salary})$ as the main response in regression analysis to improve symmetry and stabilise variance.

We first fitted a baseline linear regression on the training set using Logged_Salary as the response and all available predictors (excluding Player and Salary).

Diagnostic

Baseline diagnostic plots show that the full linear model captures the general correlation between performance and salary, but some key problems remain (Figure 4). The Residuals vs Fitted plot has a small curve, which suggests the relationship is not fully linear (the model form may be slightly wrong). The Scale-Location plot is not flat, which means error size changes across salary levels (heteroskedasticity issue). The Q-Q plot deviates away from the straight line at both ends, suggesting the presence of outliers and heavy-tailed residuals, which is likely due to a few very high-salary data points, driven by factors beyond performance. The Residuals vs Leverage plot shows a small number of high-leverage points, which means a few observations could strongly affect the fitted coefficients. Consequently, these issues motivate our targeted refinements such as variable transformations, removing highly redundant predictors, and checking (& controlling for) influential points to make coefficients more stable and easier to interpret.

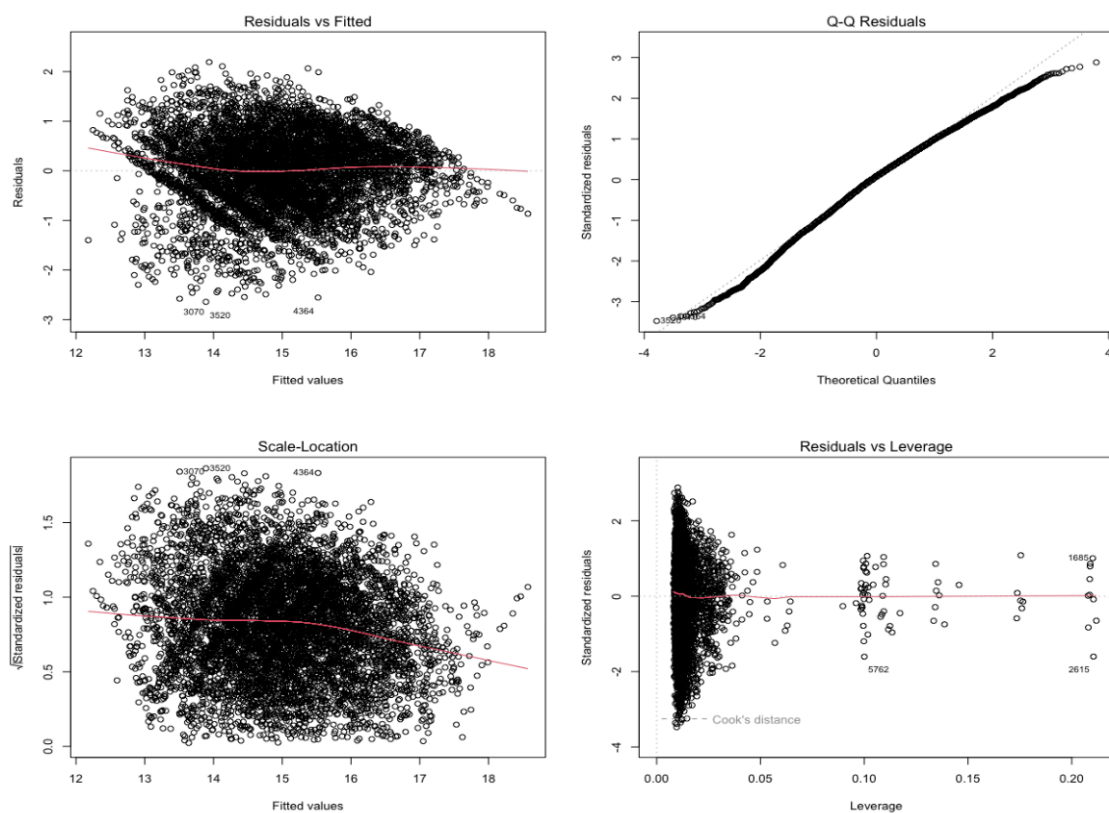


Figure 4

Model refinement (selection, collinearity, and diagnostics)

We implement the following model development steps used to obtain a stable and interpretable model:

- Variable redundancy (AIC stepwise selection): We applied stepwise selection based on AIC ($k = 2$) as an initial screening tool. We used AIC rather than BIC because our primary goal is prediction and AIC is less strict, reducing the risk of dropping useful predictors.

- Collinearity control (VIF/GVIF): We assessed multicollinearity using VIF/GVIF. Several “volume” variables were highly correlated (e.g., made vs attempts), which can “inflate” standard errors and make coefficient interpretation unstable. We therefore removed highly redundant predictors while keeping representative variables (e.g., retaining Field_Goal_Attempts while removing other overlapping makes/totals).
- Functional transformations (improving linearity and robustness): Diagnostic diagram patterns suggest non-linearity and sensitivity to extreme values. We therefore applied log transformations to selected skewed predictors (Field_Goal_Attempts, Free_Throws_Attempts, Assists, Blocks, and Defensive_Rebounds) and refitted the model to improve the linear fit and reduce the influence of extreme observations.
- Influence diagnostics (training-only removal): We computed Cook’s distance and removed influential training cases using the standard $4/n$ rule (Figure 5). This removed 378 observations from the training set ($6,934 \rightarrow 6,556$), while the test set was left unchanged to reflect a realistic prediction setting. This operation improves our RMSE value from 5,038,647 to 4,738,404, which demonstrates a substantial gain in predictive generalizability. By excluding these high-leverage points from the training process, we prevented the model from overfitting to anomalies (e.g., superstars & rookies) that deviate from standard market valuation. The performance improvement on the unprocessed test set confirms that the removed observations indeed generate statistical noise rather than valuable signals; the model was able to better predict salaries for the general population when it is not affected by extreme training cases.

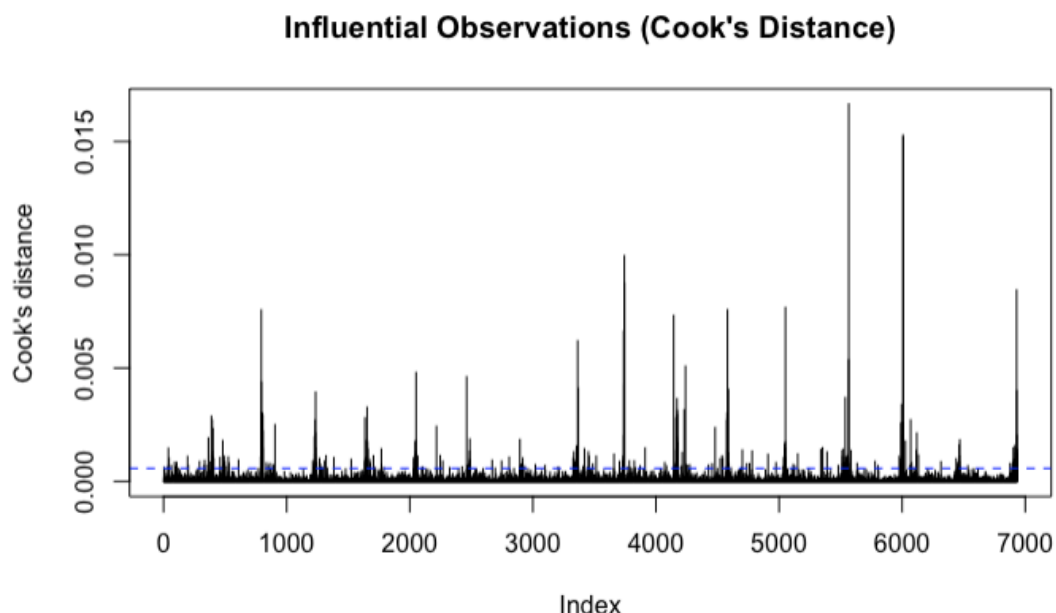


Figure 5

After these steps, the refined specification retained Year, Pos, Age, Team, Games_Played, Minutes, Field_Goal_Percent, Free_Throws_Percent, Field_Goal_Attempts, Free_Throws_Attempts, Defensive_Rebounds, Points, and selected discipline/defence measures (e.g., Assists, Blocks, Personal_Fouls), with log-transformations applied.

Final model refinement

To better reflect real-world salary patterns and lower RMSE, we made two final refinements. First, we added a quadratic age term (Age^2) because player earnings often rise to a peak and then decline, which is not well captured by a purely linear age effect. Second, we accounted for salary inflation across seasons by incorporating the NBA salary cap. Instead of modelling raw salary only, we normalised salary by cap (Salary/Cap) and modelled the log ratio, so that salaries from different years are more comparable. After predicting on the test set and converting the ratio back to dollar values, the normalised evaluation gives $\text{RMSE} = 4,669,792$ and $\text{MAE} = 2,958,477$, which is slight lower than the earlier unnormalised test errors.

Interpretation of the final model

Regression summary

Variable	Model (Estimate (SE))
PosPF-SF	0.2079 (0.2614)
PosPG	-0.5348 (0.0466)***
PosPG-SG	-0.7413 (0.2981)*
PosPF	-0.2694 (0.0371)***
PosSF-PF	-0.0235 (0.3197)
PosSF-SG	-0.0110 (0.2373)
PosSG	-0.3383 (0.0387)***
PosSG-PG	-0.5128 (0.3512)
PosSG-SF	0.0919 (0.2372)
Age	0.1738 (0.0246)***
$I(\text{Age}^2)$	-0.0019 (0.0004)***
Games_Played	0.0074 (0.0006)***
Minutes	0.0301 (0.0030)***
Points	0.0519 (0.0043)***
logAST	0.2737 (0.0407)***
logBLK	0.1352 (0.0578)*
logFTA	0.1249 (0.0419)**
Residual standard error	0.7798 (df = 6523)
Multiple R^2	0.6009
Adjusted R^2	0.5997
F-statistic	491.1000 (df = 20; 6523), $p < 2.2000 \times 10^{-16}$

Notes: Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Figure 6

Because the response is $\log(\text{Salary/Cap})$, effects can be interpreted as percentage changes; for log-transformed predictors (e.g., $\log\text{AST}$), coefficients can be read as elasticities. We demonstrate both interpretations using two examples below (see Figure 6).

Minutes ($\beta = 0.0301$, $p < 2e-16$):

“Minutes” is one of the strongest predictors in the model. Interpreting on the log ratio scale, an increase of 1 unit in Minutes is associated with about 3.06% higher predicted Salary/Cap (since $\exp(0.0301) - 1 \approx 3.06\%$), holding other variables constant.

This aligns with the idea that playing time captures both role importance and sustained trust from the team, which are strongly reflected in compensation.

Assists ($\log\text{AST}$, $\beta = 0.2737$, $p \approx 1.80e-11$):

Because “Assists” is in log form, the coefficient is an elasticity. A 10% increase in assists is associated with roughly 2.73% higher predicted Salary/Cap ($0.2737 \times 10\%$).

This interpretation also highlights the diminishing returns: with lower assist numbers, the benefit of increasing the same absolute value is greater than with already high assist numbers. This aligns with salary patterns, where higher-level players tend to receive less from additional contributions.

Non-linear variable

Our model uses Age^2 to capture the “age phenomenon” of an NBA player's career. In sport economics, the value of age does not increase in a straight line forever; it instead, in many cases, follows an inverted U-shape. If we use a simple linear “age” variable, the model might incorrectly assume that increasing age always corresponds to higher wages, which is counterfactual. Adding a squared term on age corrects this problem by transforming the regression line into a curved shape.

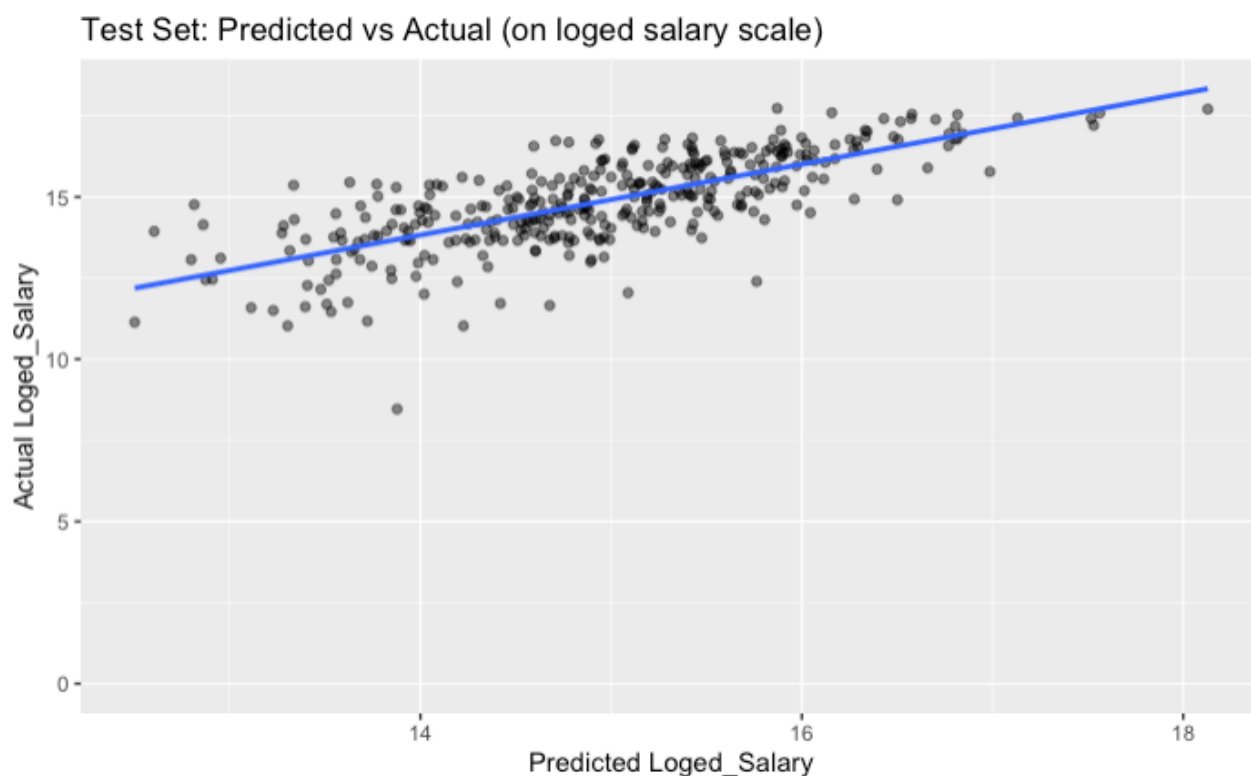
The results confirmed this dynamic. The positive coefficient for age indicates that experience generally pays off: teams reward the leadership and tactical knowledge of players. However, the negative coefficient for Age^2 acts as a “brake”. It shows that while salaries increase with age, the rate of increase gradually decreases. Therefore, the percentage increase in value gained by a rookie accumulating a year of experience is higher than the percentage increase in value gained by an experienced player accumulating the same year of experience.

Our model suggests the theoretical turning point is around 47 years old. This finding implies that within most of the observed players, the model does not predict a steep decline in value for relatively older players. Instead, it suggests an “increasing-but-flattening” trajectory where salary potential remains high but stops growing aggressively in a player's 40s.

Final Model Evaluation

Our final model consists of predictors related to player position, age, historical games played, playing time, points attained, assists, blocks, and free throw attempts. Regarding position, power forward - small forward, and shooting guard - small forward have a positive correlation with wages. All other positions more or less have a negative correlation with wages. Age has a strong and significant positive correlation with wages, alongside Games Played and Minutes, which demonstrate that player experience is positively correlated with their earnings. More performance-specific predictors: Points, Assists, Blocks, Free Throw Attempts, also show a significant positive correlation with wages, suggesting these performance statistics are the most crucial ones for wage determination.

About our model's performance on the test set (5%). In general, black dots are relatively distributed equally (Figure 7). Statistically speaking, compared to the RMSE using the full model, \$4,738,404, the RMSE of the final model is \$4,669,792, which shows some level of improvement in prediction accuracy. This indicates a moderate but realistic level of predictive accuracy given the volatile nature of NBA contracts and the intervention of superstar players. The R^2 value of our final model is around .60, which suggests a moderate level of variables being covered by our model. There are some variables that this model does not cover, which are discussed in the section "Limitation".



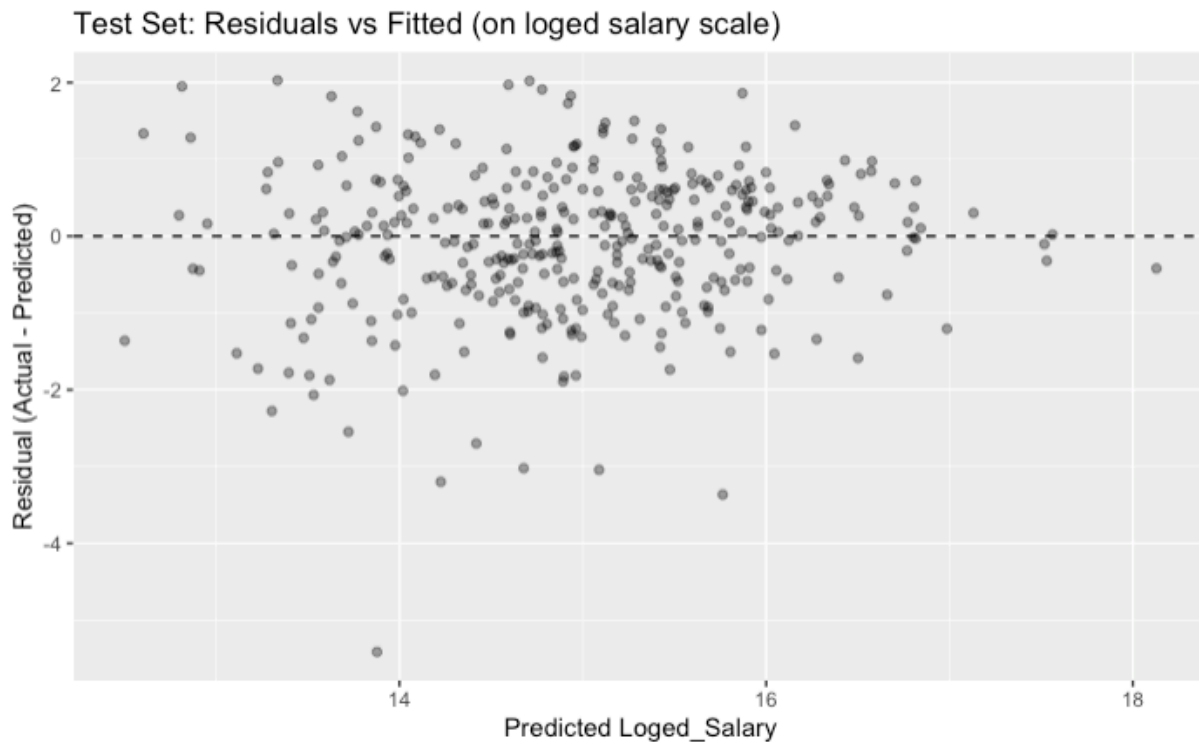


Figure 7

Limitations

Omitted variables and NBA institutional constraints

Even with detailed measures of on-court performances, NBA salaries depend on factors that our dataset does not include. Examples of this would include contract constraints, negotiation context, injury history and durability, awards, and reputation. More importantly, player contracts, especially 'max contracts', might also be determined by out-of-court marketability factors such as brand values and ticket sales, which lie beyond the scope of purely on-court statistics. These omitted factors can create large gaps between salary and what box-score performance would predict. This is especially true at the top end of the salary distribution, where these factors significantly contribute to high prediction error.

In the training set, we remove outliers, which are generally players who earn significantly higher wages. However, when applying the trained model to the test set where "outlier players" exist, our model would typically underestimate their payment because these out-of-court measurements are not included as input for our model construction. To address that, we may need to find a way to include more available variables and quantify players' off-court values and influences.

Context effects are not fully captured

At the same time, to enhance comparability between different seasons or years, our final model standardizes the target variable using the salary cap ratio (Salary / Salary Cap), effectively

eliminating the effects of inflation. However, we also recognize significant contextual differences between teams. Factors such as the purchasing power of teams, the status of championship contenders, and specific roster composition strategies create unobservable heterogeneity. This is often referred to as "team effects". A standard solution would be adding dummy variables for each team (e.g., `is_Lakers`, `is_Knicks`, etc.) to control for these time-invariant team characteristics.

However, we did not adopt this for maintaining model simplicity. Introducing these additional categorical predictors would significantly increase the model's dimensionality, potentially leading to overfitting and obscuring the interpretability of our primary objective of building this model: improving market transparency and providing information to players and teams. For our target stakeholders, "team effects" of each NBA team have little practical value. Players and agents are primarily concerned with maximizing the value of their transferable skills - variables they can control, rather than analyzing league-wide salary gaps between teams. Therefore, a model focused on the general price of talent is more relevant to career development and negotiation than one diluted by team-specific heterogeneity.

Repeated observations (panel structure)

The dataset includes repeated player-seasons for the same players. Our data has 7298 raw entries from 1658 players, which means some players will have multiple entries across years. This can create within-player correlation in errors, and it violates the independence assumption of OLS. Our main objective is prediction, not causal inference. However, this dependence can still bring several issues. Specifically speaking, treating repeated observations as independent data artificially "inflates" the sample size, leading to an underestimation of the s.d. Therefore, reported p-values may appear "too optimistic" (more Type I error - False Positive error), suggesting statistical significance, when in fact it may not exist.

Validation Constraints

Another limitation stems from the sparsity of the distribution of our categorical predictor variables. Some predictor levels appear very infrequently in the dataset, which would create structural fragility when randomly splitting into training and test sets. In some data split cases, the training set fails to capture every unique level of some categorical variables, resulting in "unseen levels" in the test set. This discrepancy can cause the prediction algorithm to fail because the regression model cannot assign coefficients to historically unobserved classes. Therefore, this instability significantly limits our ability to perform robust iterative cross-validation (e.g., k-fold cross-validation) to generate a robust RMSE value. As a result, our performance validation relies on a static hold-out set rather than a more comprehensive, averaged error metric, potentially masking variability in the model's predictive stability across different data subsets.

Conclusion

Across the modelling process, we used stepwise selection, addressed multicollinearity, applied log transformations, introduced Age², and added salary-cap normalisation. However, test-set errors remain around \$5M, which is likely partly driven by the nature of the NBA: player salaries are highly right-skewed, and top players can receive “max contracts” and marketability premiums that are not fully explained by box-score statistics. We also removed influential points only in training (to stabilise fitting) while keeping the test set raw to reflect a real prediction setting, which makes test evaluation more challenging. Finally, we observed that adjusted R² dropped compared with the full model, but this is an acceptable trade-off: the refined model removes redundant, highly collinear predictors and is more stable and interpretable, while maintaining reasonable predictive performance.

Our model aims to enhance the transparency of the NBA player market by providing information to different stakeholders. The .60 R² value suggests our model covers most of the variations, but the high RMSE (around \$5M) shows the limitation of accuracy. The first major reason is that the raw data does not include quantified “out-of-court” statistics that measure top players’ values in marketability, brand value, etc. And the second reason is that the power of “team” is not statistically represented in the raw data. Team cooperation is just as important as player individual statistics. One possible improvement to improve R² and RMSE is to accumulate more predictors, and develop a mechanism for quantifying players’ team cooperation ability and their “out-of-court” value.

Bibliography

Berri, D.J., Schmidt, M.B. and Brook, S.L. (2006) *The wages of wins: taking measure of the many myths in modern sports*. Stanford: Stanford University Press.

Rosen, S. (1981) 'The economics of superstars', *The American Economic Review*, 71(5), pp. 845-858.

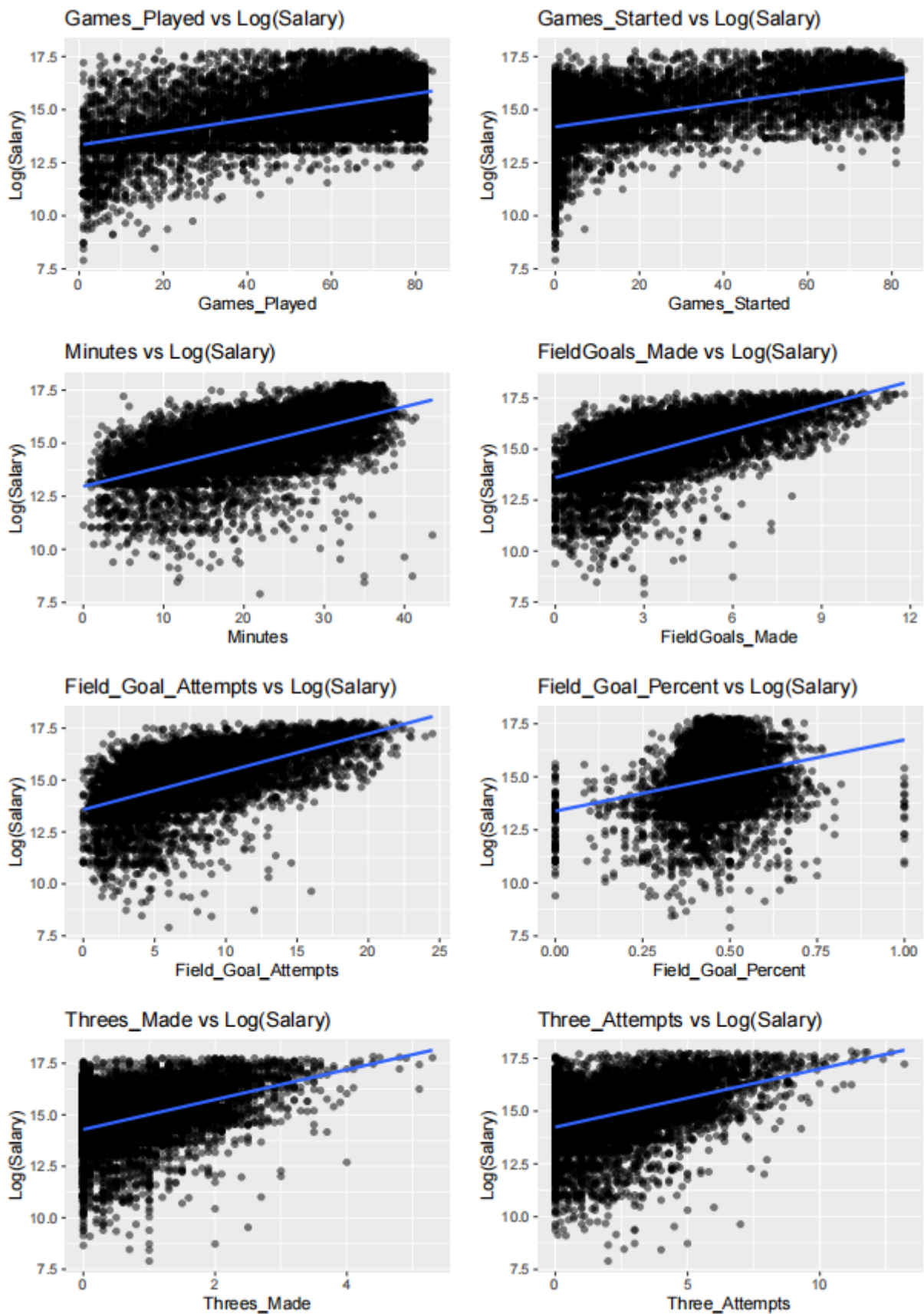
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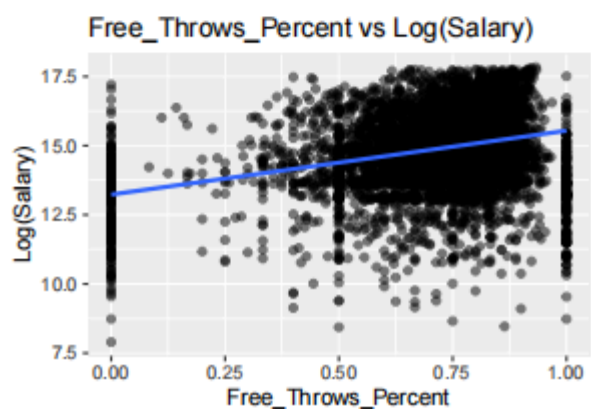
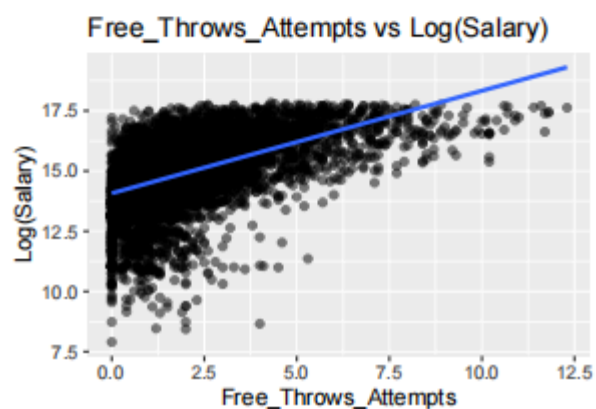
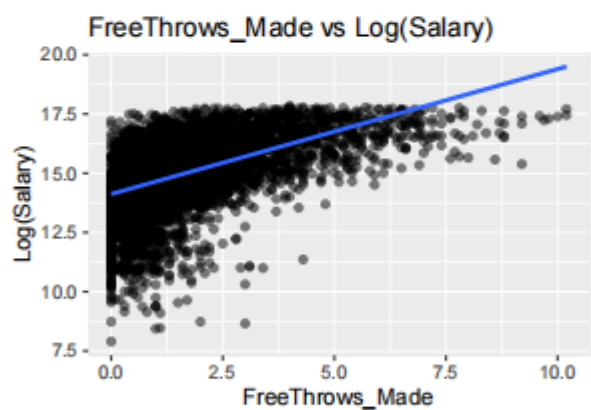
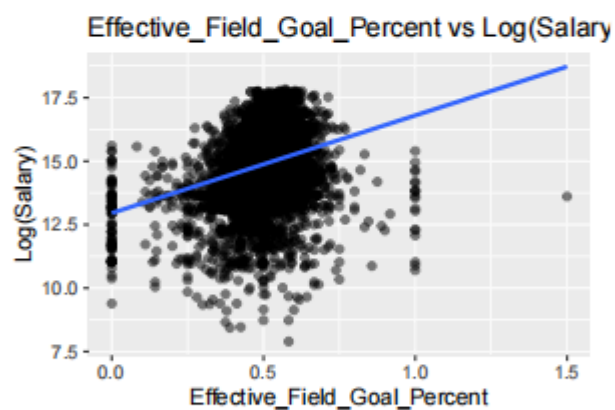
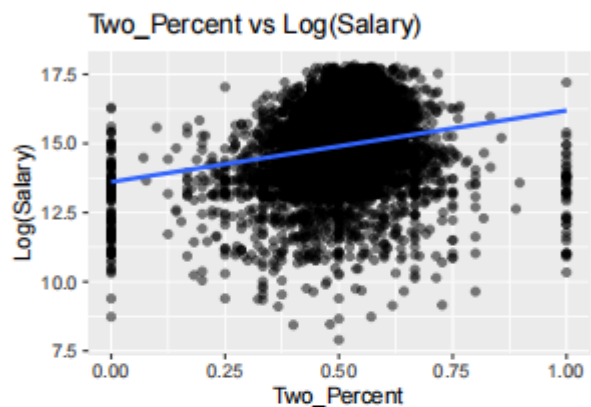
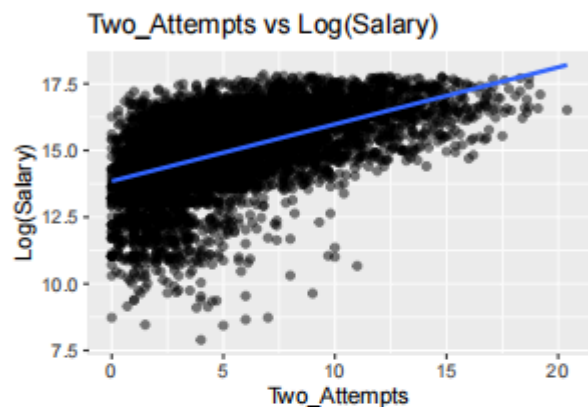
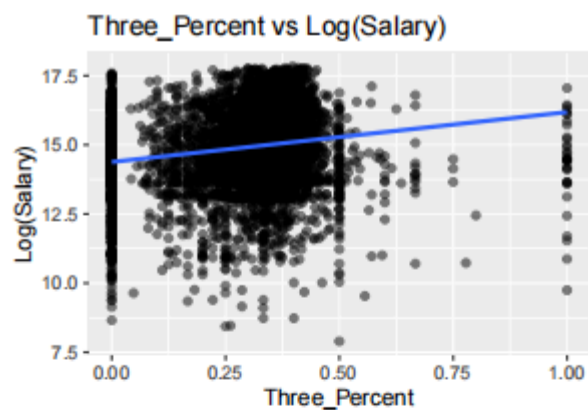
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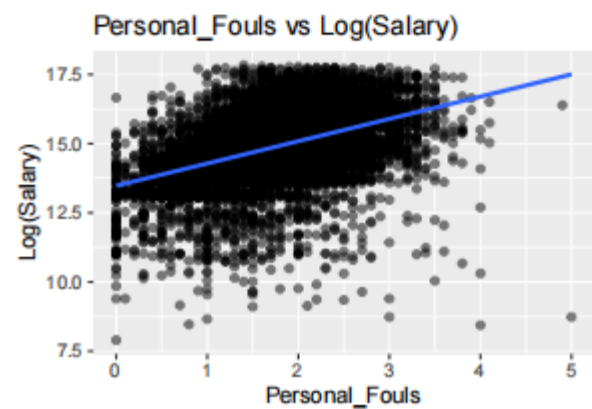
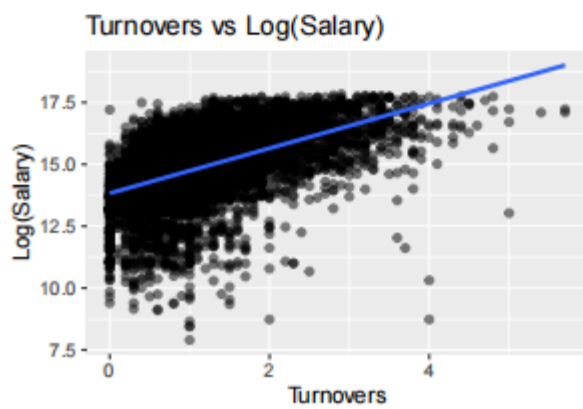
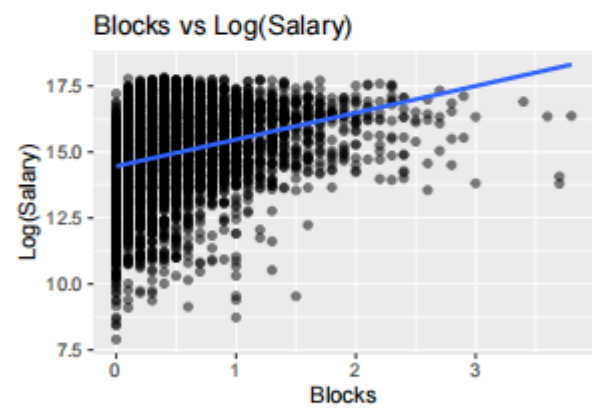
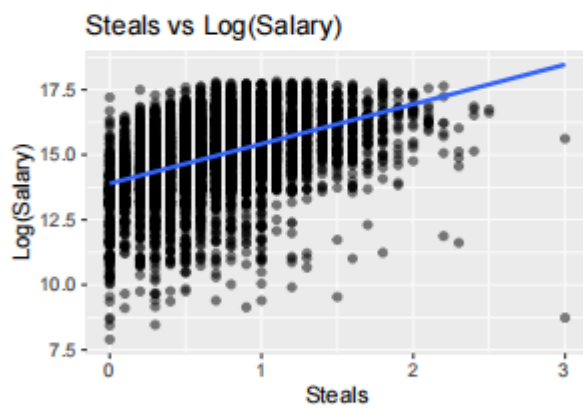
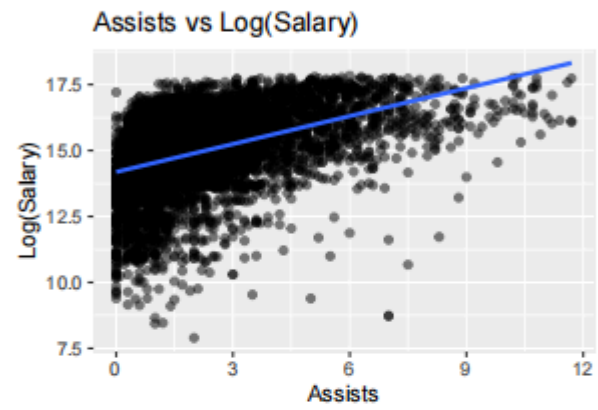
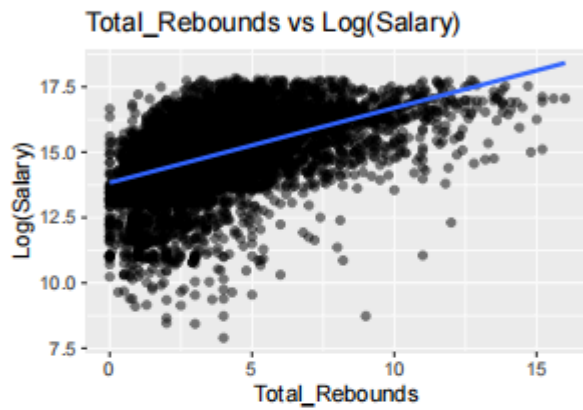
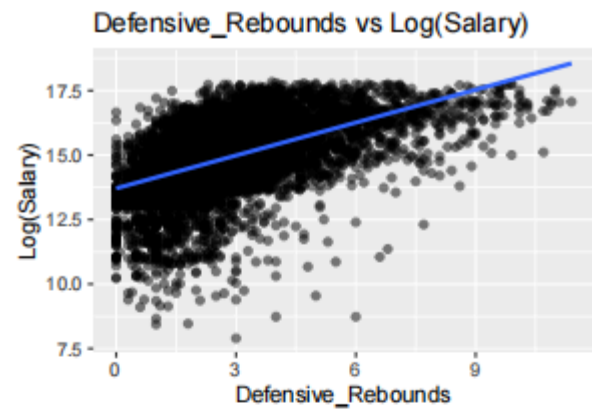
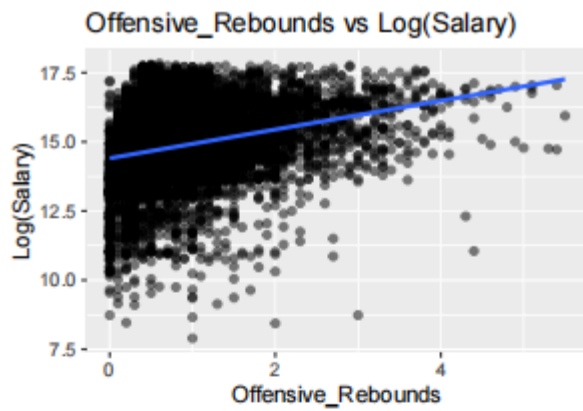
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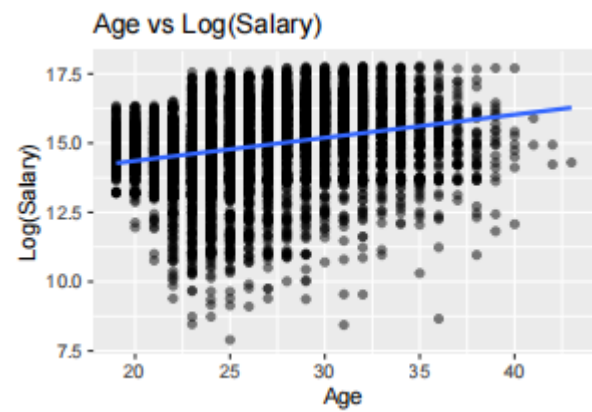
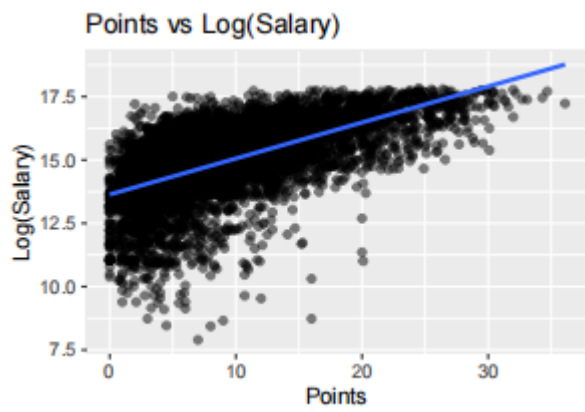
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Player	Player	Categorical	Player profile
Salary	Salary	Continuous	Outcome
Year	Year	Categorical	Player profile
Pos	Pos	Categorical	Player profile
Age	Age	Continuous	Player profile
Team	Team	Categorical	Player profile
G	Games_Played	Continuous	Performance measures
GS	Games_Started	Continuous	Performance measures
MP	Minutes	Continuous	Performance measures
FG	FieldGoals_Made	Continuous	Performance measures
FGA	Field_Goal_Attempts	Continuous	Performance measures
FG%	Field_Goal_Percent	Continuous	Performance measures
3P	Threes_Made	Continuous	Performance measures
3PA	Three_Attempts	Continuous	Performance measures
3P%	Three_Percent	Continuous	Performance measures
2P	Twos_Made	Continuous	Performance measures
2PA	Two_Attempts	Continuous	Performance measures
2P%	Two_Percent	Continuous	Performance measures
eFG%	Effective_Field_Goal_Percent	Continuous	Performance measures
FT	FreeThrows_Made	Continuous	Performance measures
FTA	Free_Throws_Attempts	Continuous	Performance measures
FT%	Free_Throws_Percent	Continuous	Performance measures
ORB	Offensive_Rebounds	Continuous	Performance measures
DRB	Defensive_Rebounds	Continuous	Performance measures
TRB	Total_Rebounds	Continuous	Performance measures
AST	Assists	Continuous	Performance measures
STL	Steals	Continuous	Performance measures
BLK	Blocks	Continuous	Performance measures
TOV	Turnovers	Continuous	Performance measures
PF	Personal_Fouls	Continuous	Performance measures
PTS	Points	Continuous	Performance measures

Appendix B









Appendix C

